

# The Language of Medicine

## 13<sup>th</sup> edition

Davi-Allen Chabner

# Chapter 10

## Nervous System

# Chapter Goals (Slide 1 of 2)

- Name, locate, and describe major organs of the nervous system and their functions.
- Learn nervous system combining forms and use them with suffixes and prefixes.
- Define pathologic conditions affecting the nervous system.

# Chapter Goals (Slide 2 of 2)

- Describe nervous system-related laboratory tests, clinical procedures, and abbreviations.
- Apply your new knowledge to understanding medical terms in the proper contexts, such as medical reports and records.

# Chapter 10

## Lesson 10.1

# Introduction (Slide 1 of 2)

Nervous system:

- Complex: 10 billion nerve cells
- Voluntary and involuntary functions
- Carries electrical messages
- External and internal receptors

# Introduction (Slide 2 of 2)

Nerve cells (neurons):

- Microscopic
- Collected into macroscopic nerves
- Carry electrical messages all over the body

# General Structure of the Nervous System (Slide 1 of 3)

Two major divisions:

- **Central nervous system—CNS**
  - Brain
  - Spinal cord
- **Peripheral nervous system—PNS**
  - Cranial nerves
  - Spinal nerves
  - Plexuses
  - Peripheral nerves throughout the body



# General Structure of the Nervous System (Slide 2 of 3)

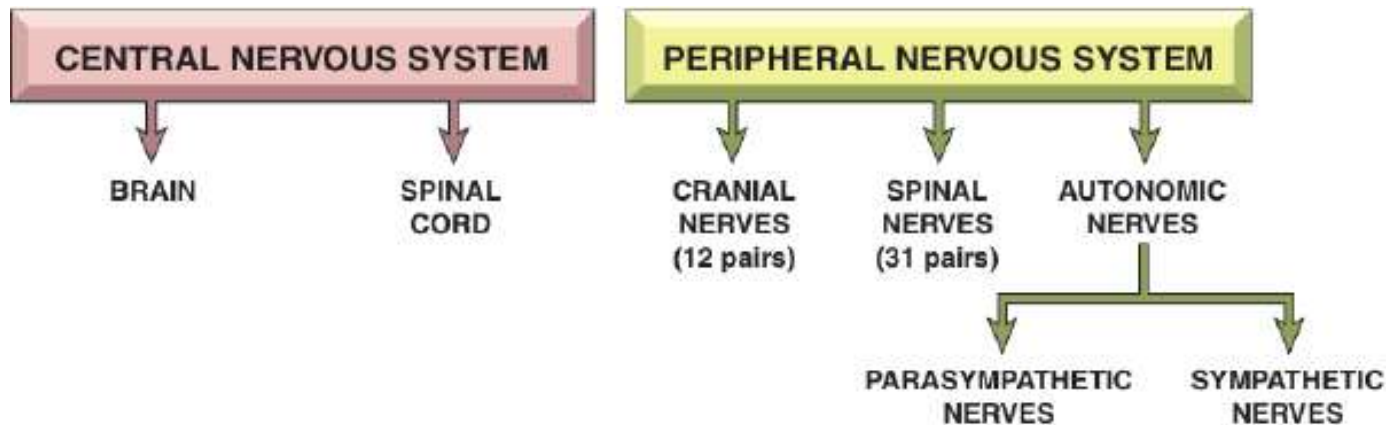
- Cranial and spinal nerves
  - Sensory nerves carry messages toward the brain
  - Motor nerves carry messages from the brain
  - Mixed nerves carry both sensory and motor fibers
- Sensory receptors

# General Structure of the Nervous System (Slide 3 of 3)

## **Autonomic nervous system:**

- Carries impulses from the central nervous system to the organs
- **Sympathetic** nerves stimulate the body under stress
- **Parasympathetic** nerves balance the sympathetic system
  - Slow heart rate
  - Lower blood pressure

# Divisions of the Central and Peripheral Nervous System



# Neurons, Nerves, and Glia (slide 1 of 5)

**Neuron** = Individual nerve cell

- Dendrite
- Cell nucleus
- Axon
  - Myelin sheath
  - Neurilemma
- Terminal end fibers (secrete neurotransmitters)
- Neurotransmitters transfer impulse across synapse

**Ganglia** = Small clusters of nerve cell bodies

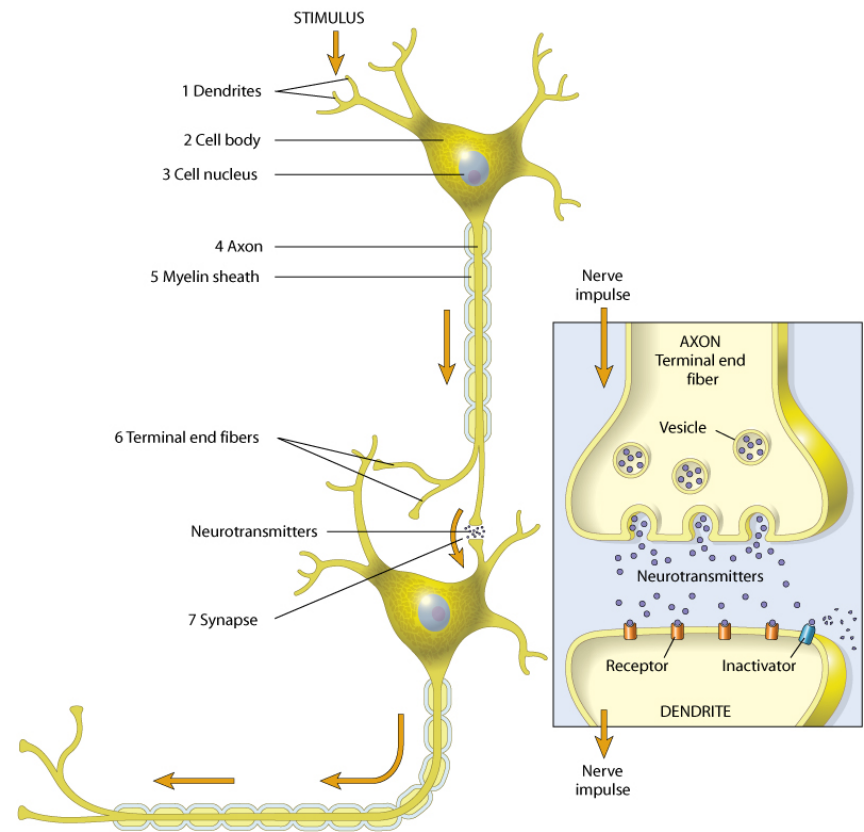
# Neurons, Nerves, and Glia (slide 2 of 5)

## **The anatomical parts of a neuron contain:**

- Dendrites
- Cell body
- Cell nucleus
- Axon
- Myelin sheath
- Terminal end fibers
- Synapse

# Neurons, Nerves, and Glia (slide 3 of 5)

- The image shows the parts of a neuron. Review the path of a nervous impulse



# Neurons, Nerves, and Glia (slide 4 of 5)

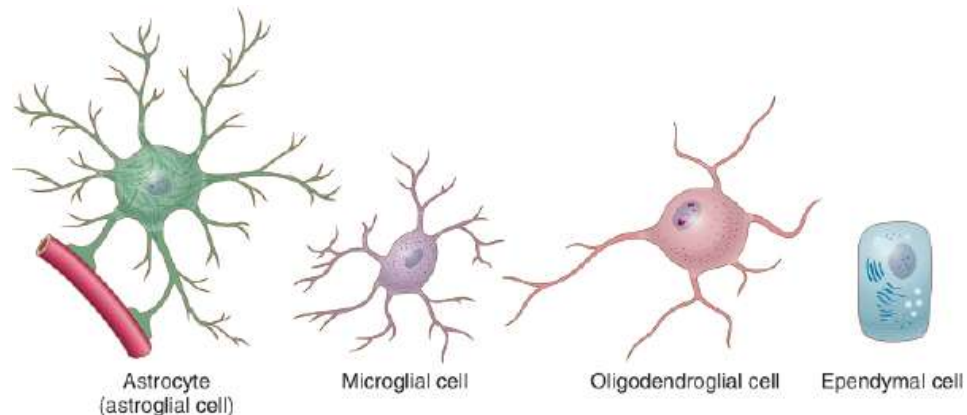
Glia cells:

- Maintain the health of the nervous system
- Do not transmit impulses

# Neurons, Nerves, and Glia (slide 5 of 5)

## Four types:

- Astrocytes (astroglial cells)
- Microglia (microglial cells)
- Oligodendroglia  
(oligodendroglial cells)
- Ependymal cells





# Chapter 10

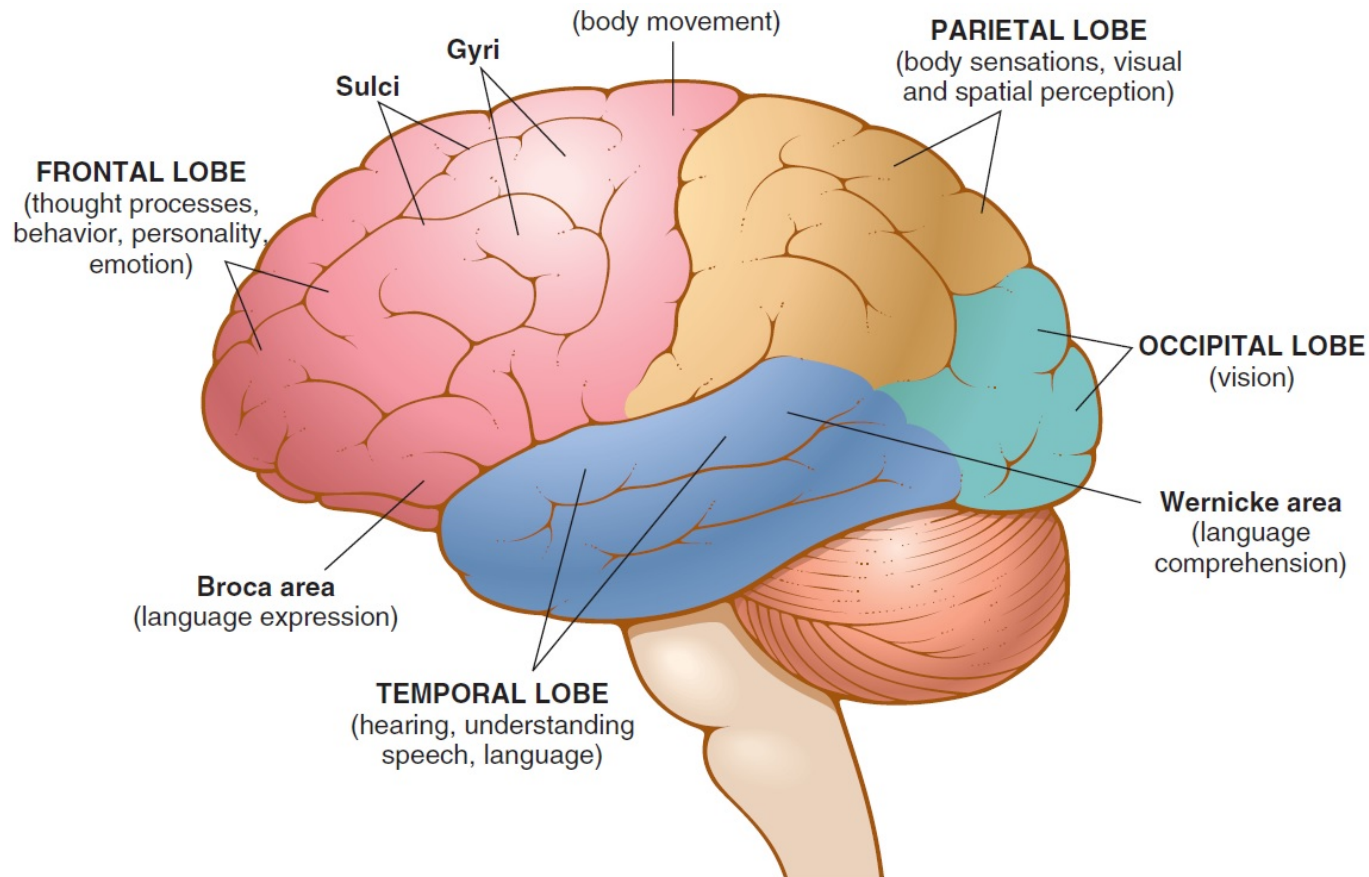
## Lesson 10.2

# The Brain (slide 1 of 7)

## **The anatomical structures of the brain include:**

- Frontal lobe
- Sulci
- Gyri
- Parietal lobe
- Occipital lobe
- Wernicke area
- Temporal lobe
- Broca area

# The Brain (slide 2 of 7)



# The Brain (slide 3 of 7)

- Cerebrum
  - The largest section of the brain
  - Surface nerve cells called cerebral cortex
  - Manages speech, vision, smell, movement, hearing, and thought

# The Brain (slide 4 of 7)

- Cerebellum
  - Coordinates voluntary movements
  - Maintains balance
- Thalamus
  - Integrates and monitors impulses from skin (pain)

# The Brain (slide 5 of 7)

- Hypothalamus
  - Controls body temperature, sleep, appetite, sexual desire, and emotions
  - Regulates release of hormones from the pituitary gland
  - Monitors sympathetic and parasympathetic nervous systems

# The Brain (slide 6 of 7)

## Brainstem

- Pons
  - Bridges the cerebrum and cerebellum with the rest of the brain
  - Houses nerves for the face and eyes
  - Midbrain
    - Uppermost portion of the brainstem
    - Contains pathways connecting the cerebrum with lower portions of brain

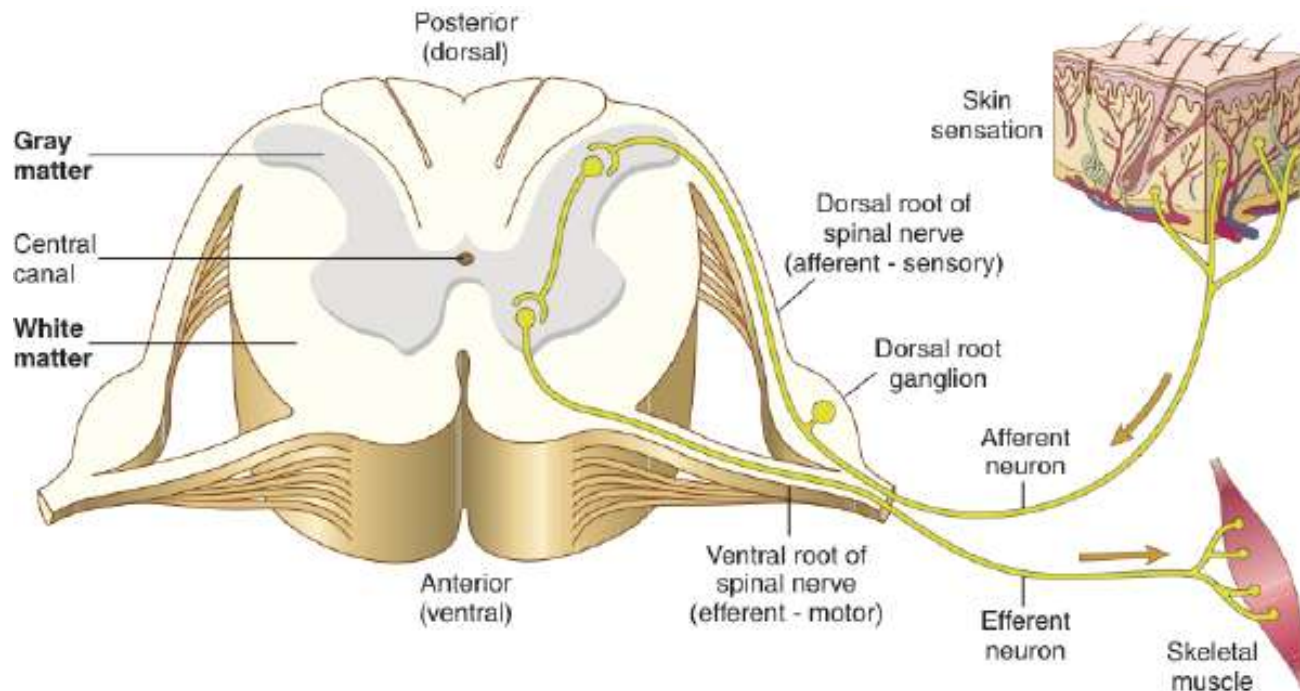
# The Brain (slide 7 of 7)

## Brainstem

- Medulla oblongata
  - Connects the spinal cord to the brain
  - Nerve tracts from side to side
  - Regulates:
    - Blood vessels
    - Heart
    - Respiratory system



# The Spinal Cord

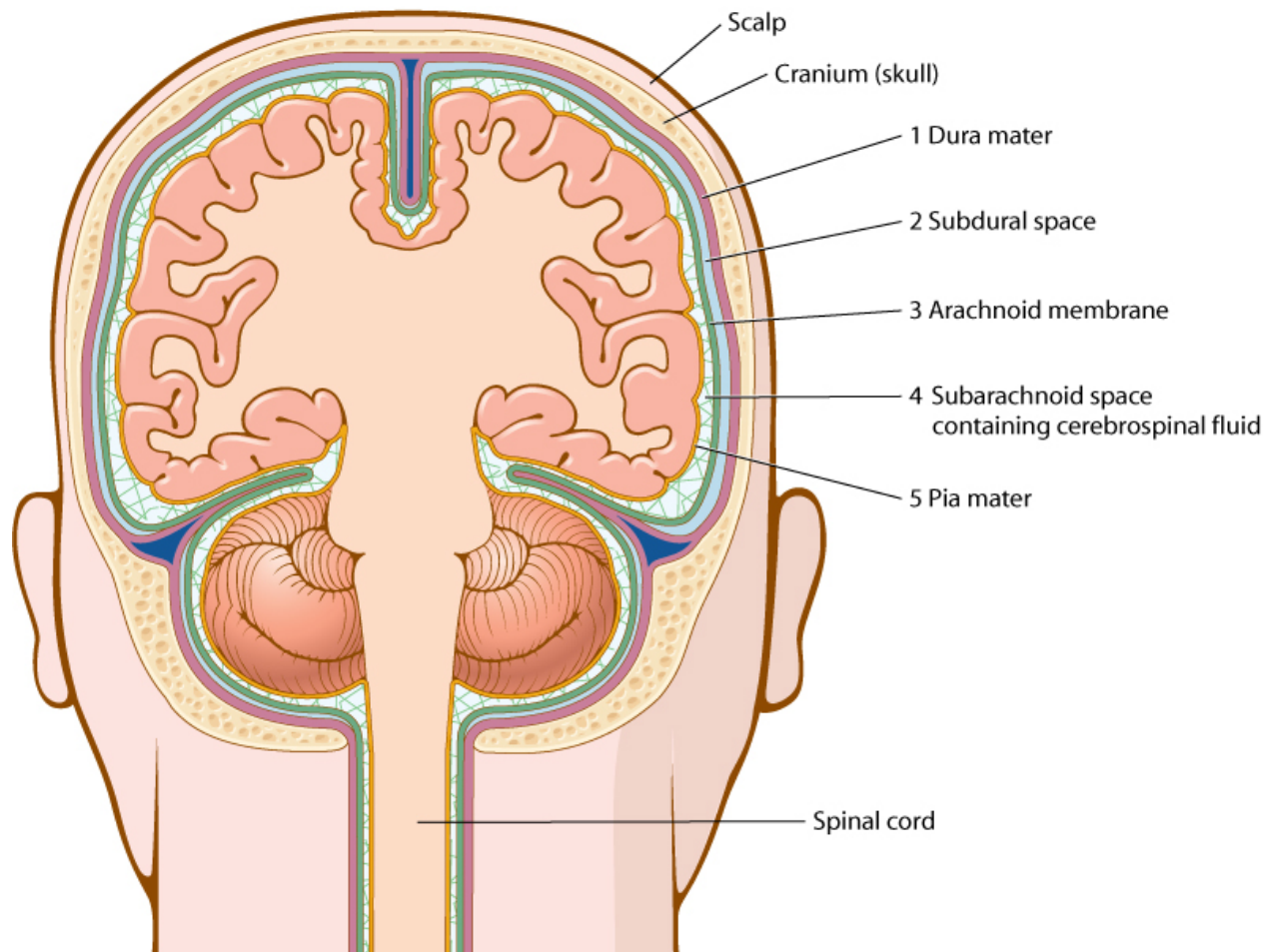


# Meninges (slide 1 of 2)

Meninges contain:

- Dura mater
- Subdural space
- Arachnoid membrane
- Subarachnoid space
- Pia mater

# Meninges (slide 2 of 2)



# Vocabulary (Slide 1 of 34)

| Term                  |
|-----------------------|
| acetylcholine         |
| afferent nerve        |
| arachnoid<br>membrane |

# Vocabulary (Slide 2 of 34)

| Term               | Meaning/Definition                                                          |
|--------------------|-----------------------------------------------------------------------------|
| acetylcholine      | Neurotransmitter chemical released at ends of nerve cells                   |
| afferent nerve     | Carries messages <i>toward</i> the brain and spinal cord                    |
| arachnoid membrane | Middle layer of the three membranes that surround the brain and spinal cord |

# Vocabulary (Slide 3 of 34)

| Term           |
|----------------|
| astrocyte      |
| autonomic      |
| nervous system |
| axon           |

# Vocabulary (Slide 4 of 34)

| Term                     | Meaning/Definition                                                                     |
|--------------------------|----------------------------------------------------------------------------------------|
| astrocyte                | Glial cell that transports water and salts from capillaries                            |
| autonomic nervous system | Nerves that control involuntary body functions of muscles, glands, and internal organs |
| axon                     | Microscopic fiber that carries the nervous impulse along a nerve cell                  |

# Vocabulary (Slide 5 of 34)

| Term                |
|---------------------|
| blood-brain barrier |
| brainstem           |
| cauda equina        |



# Vocabulary (Slide 6 of 34)

| Term                | Meaning/Definition                                                                                                                                               |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| blood-brain barrier | Protective separation between the blood and brain cells that keeps substances (such as anticancer drugs) from penetrating capillary walls and entering the brain |
| brainstem           | Posterior portion of the brain that connects the cerebrum with the spinal cord: includes the midbrain, pons, and medulla oblongata                               |
| cauda equina        | Collection of spinal nerves below the end of the spinal cord                                                                                                     |

# Vocabulary (Slide 7 of 34)

| Term                         |
|------------------------------|
| cell body                    |
| central nervous system (CNS) |
| cerebellum                   |

# Vocabulary (Slide 8 of 34)

| Term                         | Meaning/Definition                                                                      |
|------------------------------|-----------------------------------------------------------------------------------------|
| cell body                    | The part of a nerve cell that contains the nucleus                                      |
| central nervous system (CNS) | The brain and spinal cord                                                               |
| cerebellum                   | The posterior part of the brain that coordinates muscle movements and maintains balance |

# Vocabulary (Slide 9 of 34)

| Term                |
|---------------------|
| cerebral cortex     |
| cerebrospinal fluid |
| cerebrum            |

# Vocabulary (Slide 10 of 34)

| Term                | Meaning/Definition                                                                                                            |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| cerebral cortex     | Outer region of the cerebrum; contains sheets of nerve cells; gray matter                                                     |
| cerebrospinal fluid | Circulates throughout the brain and spinal cord                                                                               |
| cerebrum            | The largest part of the brain; responsible for voluntary muscle activity, vision, speech, taste, hearing, thought, and memory |

# Vocabulary (Slide 11 of 34)

| Term           |
|----------------|
| cranial nerves |
| dendrite       |
| dura mater     |

# Vocabulary (Slide 12 of 34)

| Term           | Meaning/Definition                                                                                           |
|----------------|--------------------------------------------------------------------------------------------------------------|
| cranial nerves | Twelve pairs; carry messages to and from the brain with regard to the head and neck (except the vagus nerve) |
| dendrite       | Microscopic branching fiber of a nerve cell; the first part to receive the nervous impulse                   |
| dura mater     | The thick, outermost layer of the meninges surrounding and protecting the brain and spinal cord              |

# Vocabulary (Slide 13 of 34)

| Term                                |
|-------------------------------------|
| efferent nerve                      |
| ependymal cell                      |
| ganglion ( <i>plural: ganglia</i> ) |



# Vocabulary (Slide 14 of 34)

| Term                                | Meaning/Definition                                                                                   |
|-------------------------------------|------------------------------------------------------------------------------------------------------|
| efferent nerve                      | Carries messages <i>away</i> from the brain and spinal cord; motor nerve                             |
| ependymal cell                      | Glial cell that lines the membranes within the brain and spinal cord; helps form cerebrospinal fluid |
| ganglion ( <i>plural: ganglia</i> ) | Collection of nerve cell bodies in the peripheral nervous system                                     |

# Vocabulary (Slide 15 of 34)

| Term                          |
|-------------------------------|
| glial cell (neuroglial cell)  |
| gyrus ( <i>plural: gyri</i> ) |
| hypothalamus                  |

# Vocabulary (Slide 16 of 34)

| Term                            | Meaning/Definition                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| glial cell<br>(neuroglial cell) | Supportive and connective nerve cell that does not carry nervous impulses; can reproduce itself                           |
| gyrus ( <i>plural: gyri</i> )   | Sheet of nerve cells that produces a rounded ridge on the surface of the cerebral cortex; convolution                     |
| hypothalamus                    | The portion of the brain beneath the thalamus; controls sleep, appetite, body temperature, and pituitary gland secretions |

# Vocabulary (Slide 17 of 34)

| Term              |
|-------------------|
| medulla oblongata |
| meninges          |
| microglial cell   |

# Vocabulary (Slide 18 of 34)

| Term              | Meaning/Definition                                                                                         |
|-------------------|------------------------------------------------------------------------------------------------------------|
| medulla oblongata | The part of the brain just above the spinal cord; controls breathing, heartbeat, and size of blood vessels |
| meninges          | Three protective membranes that surround the brain and spinal cord                                         |
| microglial cell   | Phagocytic glial cell that removes waste products from the central nervous system                          |

# Vocabulary (Slide 19 of 34)

| Term          |
|---------------|
| midbrain      |
| motor nerve   |
| myelin sheath |
| nerve         |

# Vocabulary (Slide 20 of 34)

| Term          | Meaning/Definition                                                                                              |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| midbrain      | Uppermost portion of the brainstem                                                                              |
| motor nerve   | Carries messages away from the brain and spinal cord to muscles and organs; efferent nerve                      |
| myelin sheath | Covering of white fatty tissue that surrounds and insulates the axon of a nerve cell; speeds impulse conduction |
| nerve         | Macroscopic cord-like collection of fibers (axons and dendrites) that carry electrical impulses                 |

# Vocabulary (Slide 21 of 34)

| Term                  |
|-----------------------|
| neuron                |
| neurotransmitter      |
| oligodendroglial cell |



# Vocabulary (Slide 22 of 34)

| Term                  | Meaning/Definition                                                                          |
|-----------------------|---------------------------------------------------------------------------------------------|
| neuron                | Nerve cell that carries impulses throughout body; parenchyma of the nervous system          |
| neurotransmitter      | Chemical messenger released at the end of a nerve cell; stimulates or inhibits another cell |
| oligodendroglial cell | Glial cell that forms the myelin sheath covering axons; oligodendrocyte                     |

# Vocabulary (Slide 23 of 34)

| Term                      |
|---------------------------|
| parasympathetic nerves    |
| parenchyma                |
| peripheral nervous system |

# Vocabulary (Slide 24 of 34)

| Term                      | Meaning/Definition                                                                                                                         |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| parasympathetic nerves    | Involuntary, autonomic nerves that regulate normal body functions such as heart rate, breathing, and muscles of the gastrointestinal tract |
| parenchyma                | Essential, distinguishing tissue of any organ or system; for the nervous system, neurons and nerves that carry impulses                    |
| peripheral nervous system | Nerves outside the brain and spinal cord; cranial, spinal, and autonomic nerves                                                            |

# Vocabulary (Slide 25 of 34)

| Term                               |
|------------------------------------|
| pia mater                          |
| plexus ( <i>plural</i> : plexuses) |
| pons                               |

# Vocabulary (Slide 26 of 34)

| Term                               | Meaning/Definition                                                                                    |
|------------------------------------|-------------------------------------------------------------------------------------------------------|
| pia mater                          | The thin, delicate inner membrane of the meninges                                                     |
| plexus ( <i>plural: plexuses</i> ) | A large, interlacing network of nerves                                                                |
| pons                               | The part of the brain anterior to the cerebellum and between the medulla and the rest of the midbrain |

# Vocabulary (Slide 27 of 34)

| Term          |
|---------------|
| receptor      |
| sciatic nerve |
| sensory nerve |

# Vocabulary (Slide 28 of 34)

| Term          | Meaning/Definition                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| receptor      | An organ that receives a nervous stimulus and passes it on to afferent nerves; the skin, ears, eyes, and taste buds are receptors |
| sciatic nerve | The nerve extending from the base of the spine down the thigh, lower leg, and foot                                                |
| sensory nerve | Carries messages toward the brain and spinal cord from receptor; afferent nerve                                                   |

# Vocabulary (Slide 29 of 34)

| Term                                |
|-------------------------------------|
| spinal nerves                       |
| stimulus ( <i>plural: stimuli</i> ) |
| stroma                              |



# Vocabulary (Slide 30 of 34)

| Term                                | Meaning/Definition                                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| spinal nerves                       | Thirty-one pairs arising from the spinal cord                                                                       |
| stimulus ( <i>plural</i> : stimuli) | An agent of change (light, sound, touch, pressure, pain) in internal or external environment that evokes a response |
| stroma                              | Connective and supporting tissue of an organ; glial cells are the stromal tissue of the brain                       |

# Vocabulary (Slide 31 of 34)

| Term                            |
|---------------------------------|
| sulcus ( <i>plural</i> : sulci) |
| sympathetic nerves              |
| synapse                         |

# Vocabulary (Slide 32 of 34)

| Term                            | Meaning/Definition                                                                                                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------|
| sulcus ( <i>plural: sulci</i> ) | Depression or groove in the surface of the cerebral cortex; fissure                                                  |
| sympathetic nerves              | Autonomic nerves that influence bodily functions involuntarily in times of stress                                    |
| synapse                         | The space through which a nervous impulse travels between nerve cells or between nerve and muscle or glandular cells |

# Vocabulary (Slide 33 of 34)

| Term                    |
|-------------------------|
| thalamus                |
| vagus nerve             |
| ventricles of the brain |

# Vocabulary (Slide 34 of 34)

| Term                    | Meaning/Definition                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------|
| thalamus                | Main relay center of the brain; conducts impulses between the spinal cord and the cerebrum                |
| vagus nerve             | Tenth cranial nerve; branches reach to the larynx, trachea, bronchi, lungs, aorta, esophagus, and stomach |
| ventricles of the brain | Canals in the brain that contain cerebrospinal fluid                                                      |

# QUICK QUIZ (Slide 1 of 4)

**1. A sheet of nerve cells that produces a rounded fold on the surface of the cerebellum is the:**

- A. microglial
- B. gyrus
- C. parenchyma
- D. stroma

# QUICK QUIZ (Slide 2 of 4)

**2. The thin, delicate inner membrane of the meninges is the:**

- A. pia mater
- B. dura mater
- C. parenchyma
- D. stroma

# Terminology – Organs and Structures

## (Slide 1 of 3)

| Combining Form    | Meaning     |
|-------------------|-------------|
| <b>cerebell/o</b> | cerebellum  |
| <b>cerebr/o</b>   | cerebrum    |
| <b>dur/o</b>      | dura mater  |
| <b>encephal/o</b> | brain       |
| <b>gli/o</b>      | glial cells |



# Terminology – Organs and Structures

## (Slide 2 of 3)

| Combining Form         | Meaning             |
|------------------------|---------------------|
| lept/o                 | thin, slender       |
| mening/o,<br>meningi/o | membranes, meninges |
| my/o                   | muscle              |
| myel/o                 | spinal cord         |
| neur/o                 | nerve               |

# Terminology – Organs and Structures

## (Slide 3 of 3)

| Combining Form   | Meaning                     |
|------------------|-----------------------------|
| <b>pont/o</b>    | pons                        |
| <b>radicul/o</b> | nerve root                  |
| <b>thalam/o</b>  | thalamus                    |
| <b>thec/o</b>    | sheath (refers to meninges) |
| <b>vag/o</b>     | vagus nerve                 |

# Terminology – Signs and Symptoms (Slide 1 of 3)

| Combining Form or Suffix    | Meaning                       |
|-----------------------------|-------------------------------|
| <b>algēs/o, -algēsia</b>    | excessive sensitivity to pain |
| <b>-algia</b>               | pain                          |
| <b>caus/o</b>               | burning                       |
| <b>comat/o</b>              | deep sleep                    |
| <b>esthesi/o, -esthesia</b> | feeling, nervous sensation    |
| <b>kines/o, -kinesia</b>    | movement                      |

# Terminology – Signs and Symptoms (Slide 2 of 3)

| Combining Form or Suffix | Meaning          |
|--------------------------|------------------|
| <b>-kinesis, kinetic</b> | movement         |
| <b>-lepsy</b>            | seizure          |
| <b>lex/o</b>             | word, phrase     |
| <b>-paresis</b>          | slight paralysis |
| <b>-phasia</b>           | speech           |

# Terminology – Signs and Symptoms (Slide 3 of 3)

| Combining Form or Suffix | Meaning             |
|--------------------------|---------------------|
| <b>-plegia</b>           | paralysis           |
| <b>-praxia</b>           | action              |
| <b>-sthenia</b>          | strength            |
| <b>syncop/o</b>          | cut off             |
| <b>tax/o</b>             | order, coordination |

# QUICK QUIZ (Slide 3 of 4)

**3. Which term means nerve pain?**

- A. Cephalgia
- B. Analgesia
- C. Hypalgesia
- D. Neuralgia

# QUICK QUIZ (Slide 4 of 4)

**4. Which type of hematoma occurs between the skull and the dura as a result of a ruptured meningeal artery, usually after a fracture of the skull?**

- A. Subdural
- B. Epidural
- C. Intracerebral

# Chapter 10

## Lesson 10.3



# Pathology

## **Congenital disorders**

- Hydrocephalus
- Spina bifida
  - Spina bifida cystica
  - Spina bifida occulta

# Degenerative Movement, and Seizure Disorders

- Alzheimer disease (AD)
- Amyotrophic lateral sclerosis (ALS)
- Epilepsy
- Huntington disease (Huntington chorea)
- Multiple sclerosis (MS)
- Myasthenia gravis (MG)
- Palsy
- Parkinson disease
- Tourette syndrome

# Infectious Disorders

- Herpes zoster (shingles)
- Meningitis
- Human immunodeficiency virus (HIV) encephalopathy

# Traumatic Disorders

- **Cerebral concussion:**

- Type of traumatic brain injury caused by a blow to the head
- Usually no evidence of structural damage to brain tissue
- Rest is very important to allow the brain to heal

- **Cerebral contusion:**

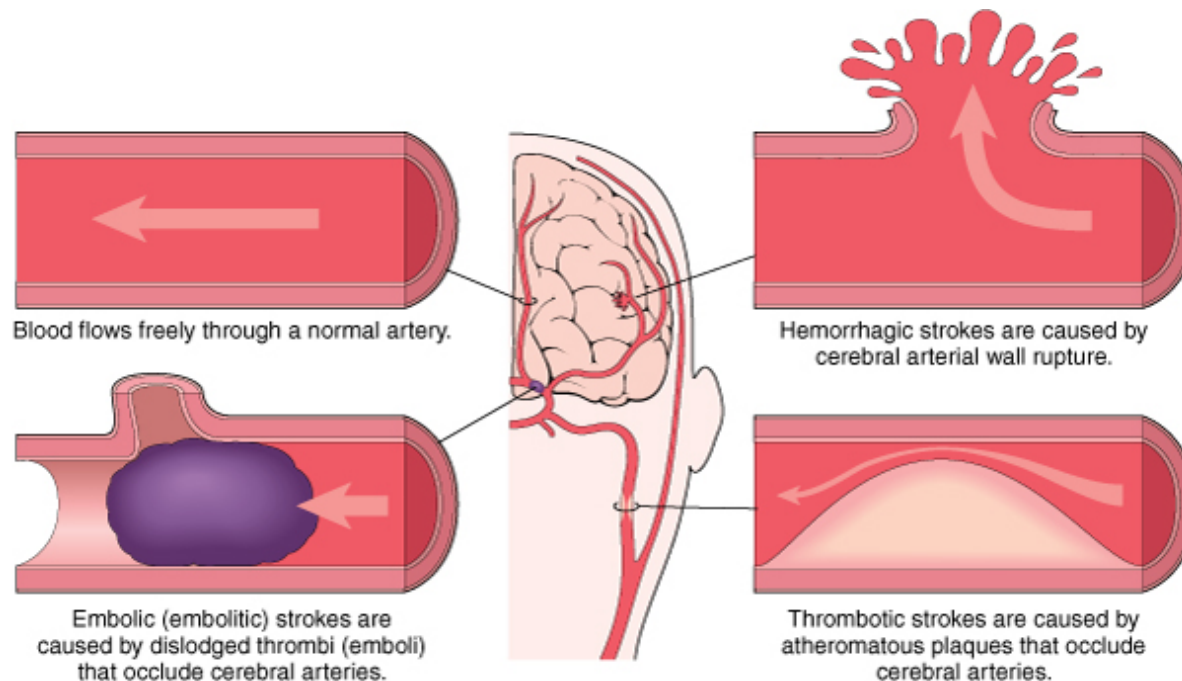
- Bruising of brain tissue as a result of direct trauma to the head
- May be associated with a fracture of the skull, edema and increased intracranial pressure
- Subdural and epidural hematomas can occur leading to permanent brain injury

# Vascular Disorders (slide 1 of 2)

- **Cerebrovascular accident (CVA)**
  - **Thrombotic** — blood clot in arteries leaving the brain
  - **Embolic** — dislodged blood clot travels to cerebral arteries
  - **Hemorrhagic** — blood vessel breaks and bleeding occurs

# Vascular Disorders (slide 2 of 2)

- **Cerebrovascular accident (CVA)**



# Clinical Procedures

## **X-ray tests:**

- Cerebral angiography
- Computed tomography (CT) of the brain

## **Other procedures:**

- MRI—Magnetic resonance Imaging
- PET—Positron emission scan
- EEG—Electroencephalography
- Doppler ultrasound studies

# Abbreviations (Slide 1 of 3)

| Abbreviation | Meaning                       |
|--------------|-------------------------------|
| AD           | Alzheimer disease             |
| AFP          | alpha-fetoprotein             |
| ALS          | amyotrophic lateral sclerosis |
| AVM          | arteriovenous malformation    |
| BBB          | blood brain barrier           |
| CNS          | central nervous system        |
| CT           | computed tomography           |
| CSF          | cerebrospinal fluid           |
| CVA          | cerebrovascular accident      |



# Abbreviations (Slide 2 of 3)

| Abbreviation | Meaning                                                |
|--------------|--------------------------------------------------------|
| EEG          | electroencephalogram                                   |
| GABA         | gamma-aminobutyric acid (neurotransmitter)             |
| ICP          | intracranial pressure; normal pressure is 5 to 15 mmHG |
| LP           | lumbar puncture                                        |
| MAC          | monitored anesthetic care                              |
| MG           | myasthenia gravis                                      |
| MRA          | magnetic resonance angiography                         |
| MRI          | magnetic resonance imaging                             |

# Abbreviations (Slide 3 of 3)

| Abbreviation | Meaning                                     |
|--------------|---------------------------------------------|
| 1/2 P        | hemiparesis                                 |
| PCA          | patient-controlled anesthesia               |
| PNS          | peripheral nervous system                   |
| PET          | positron emission tomography                |
| Sz           | seizure                                     |
| TBI          | traumatic brain injury                      |
| TENS         | transcutaneous electrical nerve stimulation |
| TIA          | transient ischemic attack                   |
| TLE          | temporal lobe epilepsy                      |
| tPA          | tissue plasminogen activator                |

# Practical Applications

- What symptoms signal nervous system problems?
- What tests are conducted to diagnose these problems?
- Why are an MRI and an MRA ordered?
- What are common pathologies of the nervous system?

# Review Sheet (Slide 1 of 5)

| Combining Form    | Meaning             |
|-------------------|---------------------|
| <b>algēs/o</b>    | sensitivity to pain |
| <b>angi/o</b>     | vessel, blood       |
| <b>caus/o</b>     | burn, burning       |
| <b>cephal/o</b>   | head                |
| <b>cerebell/o</b> | cerebellum          |
| <b>cerebr/o</b>   | cerebrum            |

# Review Sheet (Slide 2 of 5)

| Combining Form    | Meaning           |
|-------------------|-------------------|
| <b>comat/o</b>    | deep sleep        |
| <b>crani/o</b>    | skull             |
| <b>cry/o</b>      | cold              |
| <b>dur/o</b>      | dura mater        |
| <b>encephal/o</b> | brain             |
| <b>esthesi/o</b>  | nervous sensation |

# Review Sheet (Slide 3 of 5)

| Combining Form             | Meaning             |
|----------------------------|---------------------|
| <b>gli/o</b>               | glue                |
| <b>hydr/o</b>              | water               |
| <b>kines/o</b>             | movement            |
| <b>lept/o</b>              | thin, slender       |
| <b>lex/o</b>               | word, phrase        |
| <b>mening/o, meningi/o</b> | membranes, meninges |
| <b>my/o</b>                | muscle              |

# Review Sheet (Slide 4 of 5)

| Combining Form | Meaning     |
|----------------|-------------|
| myel/o         | spinal cord |
| narc/o         | sleep       |
| neur/o         | nerve       |
| olig/o         | scanty      |
| pont/o         | pons        |
| radicul/o      | nerve root  |

# Review Sheet (Slide 5 of 5)

| Combining Form  | Meaning                     |
|-----------------|-----------------------------|
| <b>spin/o</b>   | spine                       |
| <b>syncop/o</b> | cut off                     |
| <b>tax/o</b>    | order                       |
| <b>thalam/o</b> | thalamus                    |
| <b>thec/o</b>   | sheath (refers to meninges) |
| <b>troph/o</b>  | nourishment                 |
| <b>vag/o</b>    | vagus nerve                 |